Survey-Scale Galaxy Chirality with Equivariant TTA: A -0.122σ Subsample-Mask $\ell = 1$ Null, a Quantifiable Monopole-Mask Leakage Channel, and Diagnostic Evidence for a Depth/Morphology-Correlated Canonical-Mask Residual on 8.47 Million DESI Legacy Galaxies (3.2 Million Spirals)

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We report a multi-survey, equivariance-corrected angular dipole analysis of **3,201,160** DESI Legacy spiral galaxies (8.47 M sources, 471 049 high-confidence per-spiral after $p_{\rm CW}^{\rm eq} > 0.9$). The headline scientific result is a **null** $\ell = 1$ chirality-dipole observable on the analysis subsample mask: the MASTER-deconvolved single-mode pseudo-C_1 on the strict-superset subsample mask ($n = 5{,}547{,}858$, $f_{\rm sky} = 0.659$) yields -0.122σ *(500-MC label-shuffle null)*, consistent with no dipole at $\ell = 1$. The real-space post-TTA Catalog C dipole is $+0.43\sigma$ *($p = 0.30$, per-pixel-shuffle null).* Note: σ values throughout this paper are defined relative to their respective null procedures and are not directly comparable across estimators; see Table II for the mapping of each result to its null. We emphasize at the outset that this $\ell = 1$ observable is the *isotropy-breaking axial-vector channel* and is *parity-EVEN*: it is NOT a direct parity-violation test (the parity-odd analog requires 3D spin-vector or polarization-rotation cross-correlation observables outside this paper's scope). Prior literature has at times conflated these two channels; we keep them strictly separated.

A canonical-mask diagnostic of the leakage mechanism shows that pre-MASTER raw pseudo-C_1 in the un-monopole-subtracted CW-fraction map ($f_{\rm sky} = 0.49005$, $N_{\rm spiral} = 3{,}201{,}160$) is reproduced at 99.3% of its observed amplitude by a controlled monopole-only generative null ($N = 500$, binomial realizations at $p_{\rm CW}^{\rm global} = 0.4974$ on the canonical mask): a small uniform CW-vs-CCW classifier monopole couples through patchy survey-mask geometry to inflate the raw pseudo-C_ℓ at $\ell = 1$, and MASTER mode-coupling deconvolution removes the leakage. The post-MASTER canonical-mask direct-MC residual is $+3.64\sigma$ *($z = \Delta/\sigma_{\rm null}$ moment-ratio; empirical rank $p_{\rm MC} = 0.030$, i.e. $\approx 1.9\sigma$ Gaussian-equivalent; 500-MC binomial per-pixel-shuffle null)* under proper galaxy-weighted monopole subtraction. The $+3.64\sigma$ canonical-mask residual is consistent with monopole leakage through survey geometry (Sec. IV D) and is *not* interpreted as a cosmological signal.

Three interpretations of the canonical-mask residual are systematically tested with a four-null battery + direct cross-spectrum: (i) a clean real cosmological dipole at amplitude $\sim 1.7\%$, (ii) a coherent depth/sampling-correlated systematic at low ℓ on the patchy canonical footprint, or (iii) a NaMaster low-ℓ deconvolution artifact. Interpretation (iii) sharp-edge variant is disfavored by the apodized-mask test ($+3.57\sigma$ on C^2 2° apodization, essentially unchanged from binary). Interpretation (i) as a clean dipole-only explanation is disfavored by three independent lines of evidence: (a) $\ell = 2 > \ell = 1$ broadband structure; (b) absence of monotonic $p_{\rm eq}$ quality-quartile scaling; (c) direct cross-spectrum $C(A_p \times n_{\rm total})$ at $\ell = 2$ gives $r = -0.65$ with $\sigma = -2.89$ against permutation null. Interpretation (ii) is attributed to a coherent depth/sampling-correlated systematic at low ℓ on the patchy canonical footprint. Full systematic analysis is in Appendix D.

Falsification criterion. A future survey detecting a chirality dipole at $\sigma > 5$ with full amplitude $\gtrsim 0.75\%$ (the demonstrated empirical 50%-recovery-at-3σ threshold under the adopted per-pixel-shuffle null on the HC pipeline) would falsify the present null. **Scope.** The $\ell = 1$ subsample-mask null is the primary scientific result; the canonical-mask residual is interpretation (ii) systematic, not a primordial detection. The catalog (3.2 M spirals), model weights, and all reproducibility scripts are publicly released at the project repository.

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CONTENTS

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I. INTRODUCTION

The handedness (chirality) of spiral galaxies—whether their arms trail clockwise (CW) or counter-clockwise (CCW) as projected on the sky—is a simple observable that, under the trailing-arm assumption and in the absence of confounding selection effects, traces the angular-momentum direction of each disk galaxy. Throughout this paper, "CW/CCW" refers to the *projected apparent arm-winding chirality*, not a deprojected 3D spin vector. In a statistically isotropic and parity-symmetric universe, the CW and CCW fractions should be exactly equal when averaged over large angular scales. A significant directional departure would constrain isotropy-breaking axial-vector sectors in the galaxy-formation chain. The present paper is a standalone observational result: our null dipole at sub-percent sensitivity does not depend on any unpublished companion work.

Claims of such a signal have appeared intermittently in the literature. Shamir (2012) [4] reported a 2–4σ dipole with per-bin asymmetry amplitudes of ~ 5–20% using $\sim 1.27\times10^5$ SDSS galaxies. Shamir (2020) [1] and Shamir (2022) [3] reported results with ~ 2–4% asymmetries on DESI Legacy samples ("nearly 1.3×10^6 spiral galaxies" per the published abstract). Iye *et al.* (2021) [5] analyzed Galaxy Zoo data and found no significant signal after correcting for reading-direction bias and documented photometric-object duplication in earlier Shamir catalogs. Tadaki *et al.* [6] likewise found null results. Jia *et al.* [7] introduced CE-ResNet, a chirality-equivariant CNN guaranteeing by construction that flipping an input exactly swaps CW and CCW outputs, yielding CW/CCW $= 0.998$ on $\sim$1.95 million galaxies.

In this paper we present a new chirality catalog advancing beyond CE-ResNet in three respects: (i) survey-scale coverage of 8.47 million galaxies (3,201,160 equivariant-classified spirals, 1.6$\times$ CE-ResNet's scale); (ii) a dedicated NOT_SPIRAL class preventing contamination from ellipticals and irregulars; and (iii) a multi-axis bias-hardening audit suite. We use this catalog to perform a chirality dipole measurement with an empirical 50%-recovery-3σ injection-recovery threshold at $|A_{\rm dipole}| \geq 0.75\%$. The measured dipole is consistent with null: the equivariant CW fraction is 0.4974 ± 0.000279 and the post-MASTER dipole significance is -0.122σ (subsample mask, headline) / $+0.43\sigma$ (real-space cross-check). This is inconsistent in amplitude with Shamir's claimed $\sim$3% signal by a factor of $\sim$6–12 under the present pipeline, though a matched-footprint Ganalyzer reanalysis is required for a likelihood-level exclusion.

II. DATA

A. Galaxy Images

Our parent sample is the Smith42/galaxies dataset on HuggingFace (`https://huggingface.co/datasets/Smith42/galaxies`), containing 8,474,688 galaxy images from the DESI Legacy Imaging Surveys DR8 [8]. Each image is a 224×224 pixel cutout in grz bands at $0.262''$/pixel. The dataset includes unique `dr8_id` identifiers; sky coordinates are obtained by cross-matching against the Galaxy Zoo DESI predictions catalog [9]. The parent-sample selection function inherits from Galaxy Zoo DESI: photometric types REX/DEV/EXP/SER, $r \leq 19.0$, half-light radius $\geq 3''$. DR8 comprises three distinct imaging campaigns: BASS+MzLS ($\delta > +32°$), DECaLS ($\delta < +32°$), and a DES overlap region.

B. Training Labels

We assemble training labels from three sources: (1) Galaxy Zoo 1 [10]: 6,637 galaxies with CW/CCW labels at $>70\%$ vote confidence; (2) CE-ResNet [7]: 17,153 galaxies with high-confidence spiral classifications; (3) Synthetic hard negatives: 2,000 artificial images as NOT_SPIRAL training examples. The combined training set contains 26,636 images (80/20 train/validation split).

Note: 67.6% of training labels derive from CE-ResNet predictions; validation metrics against the full training set therefore partially reflect agreement with CE-ResNet rather than independent ground truth. The independent GZ1 cross-match on 234,282 disjoint matches yields spiral-chirality accuracy 69.91% (Cohen's $\kappa = 0.40$). We treat 69.91% as the conservative accuracy floor and propagate it to all downstream isotropy bounds via the sub-percent systematic floor in Sec. IV C.

III. METHODS

A. Declared Analysis Hierarchy

We declare the estimator hierarchy used throughout:

- **Primary cosmological estimators:** (i) real-space CW-fraction dipole fit on Catalog C at NSIDE=64 ($\sigma_{\text{dipole}} = 0.43$, $p = 0.30$); and (ii) MASTER-deconvolved C_ℓ at $\ell = 1$ on the analysis subsample mask ($n = 5{,}547{,}858$, $f_{\text{sky}} = 0.659$; -0.122σ).
- **Secondary diagnostic estimators:** (iii) canonical-N direct-MC NaMaster at $\ell = 1$ on the patchy canonical mask ($f_{\text{sky}} = 0.49005$, $+3.64\sigma$); and (iv) hemisphere maximum-asymmetry (3.05σ local maximum; Appendix C). These characterize how a non-zero global monopole leaks through the canonical-mask geometry.
- **Generative monopole-only null:** (v) $N = 500$ binomial-monopole realizations demonstrating the $+3.64\sigma$ canonical value is consistent with monopole-mask leakage (Sec. IV D).
- **Sensitivity floor:** (vi) empirical injection-recovery on the HC-spiral subsample (Sec. VI A): 50%-recovery-at-3σ threshold at $A = 0.75\%$.

Table I consolidates per-estimator N_{spiral}, f_{sky}, mask, null type, and reported σ.

B. Model Architecture

The classifier consists of a ViT-Small encoder [12] (`vit_small_patch16_224`, ImageNet-pretrained) with the last 6 of 12 transformer blocks fine-tuned, followed by a custom classification head:

$$\text{LayerNorm} \to 384{\to}512\ (\text{GELU},\ d{=}0.3) \\ \to 512{\to}256\ (\text{GELU},\ d{=}0.2) \to 256{\to}3\ (\text{softmax}). \tag{1}$$

The three-class output (P_{CW}, P_{CCW}, P_{NS}) is essential for full-survey deployment: applying a binary classifier to data where $\sim 70\%$ of objects are elliptical or irregular produces a catalog dominated by noise. Full architecture and training details are in Appendix B.

C. Test-Time Equivariant Averaging

At inference, each galaxy is classified on both the original image and its horizontal reflection. The equivariant probability is:

$$\begin{aligned} P^{\text{eq}}_{\text{CW}} &= \tfrac{1}{2}\left(P^{\text{orig}}_{\text{CW}} + P^{\text{flip}}_{\text{CCW}}\right), \\ P^{\text{eq}}_{\text{CCW}} &= \tfrac{1}{2}\left(P^{\text{orig}}_{\text{CCW}} + P^{\text{flip}}_{\text{CW}}\right), \\ P^{\text{eq}}_{\text{NS}} &= \tfrac{1}{2}\left(P^{\text{orig}}_{\text{NS}} + P^{\text{flip}}_{\text{NS}}\right). \end{aligned} \tag{2}$$

This procedure enforces flip-equivariance of the output protocol (flip-swap correlation = 1.000). We restrict to 2-fold TTA (original + horizontal flip) rather than the full D_4 group because mirrors flip chirality by definition, whereas in-plane rotations do not change chirality; rotation-TTA probes classifier non-equivariance rather than the chirality assignment itself. A direct D_4-TTA hold-out on two independent $\sim$2,000-galaxy subsamples confirms the mean per-galaxy P_{CW} is stable under Z_2 and D_4 to within $|\Delta\langle p_{\text{CW}}\rangle| < 0.0016$; per-galaxy argmax labels flip in 21.4% of cases between Z_2 and D_4 on borderline galaxies with $P_{\text{CW}} \approx P_{\text{CCW}} \approx 0.4$. Full details in Appendix B.

D. Catalog Tiers

The pipeline produces three tiers: **Catalog A** (raw, single-pass softmax); **Catalog B** (Platt-calibrated, +0.4% excess); **Catalog C** (equivariant production, 2-fold flip TTA). **Catalog C is the recommended tier for all cosmological parity analyses.** All three tiers share 8,474,531 rows in Apache Parquet format.

IV. RESULTS

Significance conventions. This paper reports significance in standard-deviation units (σ) throughout, but values from distinct null procedures are *not* directly comparable. Section identifiers specify the null for each result.

A. Catalog Statistics

The final catalog contains 8,474,531 galaxies (157 of 8,474,688 failed quality checks). Catalog C (equivariant): CW 1,592,107 (18.78%), CCW 1,609,053 (18.99%), NS/edge-on 5,273,371 (62.23%); spiral total $N_{\text{spiral}} = 3{,}201{,}160$ (37.78%). The spiral fraction is consistent with magnitude-limited survey expectations. Mean classification confidence is 0.951, median 0.9997.

TABLE I. Headline-estimator summary. $N_{\text{catalog spiral}}$ is the underlying Catalog C spiral count; $N_{\text{map weighted}}$ is the pixel-weighted galaxy count feeding the analysis-mask map.

Estimator	$N_{\text{catalog spiral}}$	$N_{\text{map weighted}}$	f_{sky}	Mask	Null	σ
(i) real-space dipole	3,201,160	—	—	none	isotropic bootstrap ($N = 10{,}000$)	+0.43
(ii) MASTER deconv	3,201,160	5,547,858	0.659	subsample	pp-shuffle	−0.122
(iii) canonical MASTER	3,201,160	—	0.49005	canonical	pp-shuffle	+3.64
(iv) hemisphere LEE (MC)	3,201,160	—	0.49005	canonical	max-stat MC	$p_{\text{LEE}} \le 10^{-4}$
(v) monopole+mask null	3,201,160	—	0.49005	canonical	monopole-only	+1.68
(vi) injection floor	471,049 HC	—	—	—	pp-shuffle	50%-rec-3σ at $A = 0.75\%$

TABLE II. Global CW fraction across catalog tiers. Uncertainties are 1σ binomial with $N_{\text{spiral}} = 3{,}201{,}160$.

Tier	cw/(cw + ccw)	Excess (%)	Dev. (σ)
A (raw)	0.5079 ± 0.0003	+0.79	28.8
B (calibrated)	0.504 ± 0.0003	+0.4	14.6
C (equivariant)	0.4974 ± 0.0003	−0.26	9.5

B. Global CW Fraction

The Catalog C residual (9.5σ from 0.5000, Table II) is spatially uniform across 7 equatorial coordinate slabs and does not produce a dipole. This monopole offset is a classifier artifact, not a physical signal; three candidate mechanisms are (1) GZ1 training-label CW excess, (2) residual orientation-dependent bias not corrected by 2-fold TTA, and (3) photometric asymmetry in DESI Legacy imaging. The 3.86× asymmetry-suppression factor from raw +2.05% to equivariant −0.53% demonstrates the dominance of the equivariant TTA processing. Crucially, the spatial uniformity (all 7 equatorial coordinate slabs within 0.5% of 50/50; available in the companion data repository) means this monopole does not produce a dipole pattern; the sub-percent sensitivity claim should therefore be interpreted as a floor on the physical isotropy-breaking dipole signal.

C. Dipole Analysis

We pixelize the sky at HEALPix resolution `NSIDE` $= 64$ (49,152 pixels, ~ 0.84 deg^2 per pixel). In each pixel p containing > 10 spiral galaxies, we compute the asymmetry

$$A_p = \frac{N_{\text{CW}}^{(p)} - N_{\text{CCW}}^{(p)}}{N_{\text{CW}}^{(p)} + N_{\text{CCW}}^{(p)}}. \tag{3}$$

a. Simple dipole. Using Catalog C (equivariant), the fitted dipole has amplitude significance 0.43σ ($p = 0.30$ from the isotropic-null bootstrap at $N_{\text{MC}} = 10{,}000$), fully consistent with the null hypothesis. In contrast, Catalog A (raw) shows a 2.31σ real-space dipole and a $+6.48\sigma$ pre-MASTER pseudo-C_ℓ in the lowest bandpower—both entirely artifacts of the model's residual CW bias spatially modulated by non-uniform survey depth, and both collapsed to null by two complementary reductions (equivariant TTA averaging in real space and MASTER mode-coupling deconvolution in spherical-harmonic space).

b. Angular power spectrum. We perform MASTER mode-coupling deconvolution using NaMaster [32] at $N_{\text{side}} = 64$. The MASTER-deconvolved single-mode $\ell = 1$ value is $C_1^{\text{meas}} = 1.494 \times 10^{-6}$, compared with null mean $\langle C_1^{\text{null}} \rangle = 1.546 \times 10^{-6}$ ($\sigma_{\text{null}} = 4.290 \times 10^{-7}$) from 500 Monte Carlo realizations, yielding $(C_1^{\text{meas}} - \langle C_1^{\text{null}} \rangle)/\sigma_{\text{null}} = -0.122\sigma$ on the subsample mask ($f_{\text{sky}} = 0.659$). This is the primary isotropy-breaking dipole observable. NaMaster configuration details are in Appendix A.

D. Monopole+Mask Leakage Generative Null

The canonical-mask direct-MC $\ell = 1$ value of $+3.64\sigma$ and the local hemisphere maximum of 3.05σ were interpreted in earlier paper versions as mask-geometric leakage of the global 9.5σ monopole. We formalize this with a generative null: $N = 500$ realizations in which the per-pixel CW count is drawn from Binomial$(n_{\text{total}}, p_{\text{CW}}^{\text{global}})$ on the exact canonical mask, with no injected dipole.

The monopole-only null reproduces 99.3% of the observed pre-MASTER pseudo-$C_\ell^{(\ell=1)}$ power. The prior literature's pre-MASTER dipole-detection claims are therefore explained at the percent level by this leakage channel under our DESI/ViT-Small pipeline. MASTER decoupling removes the canonical-mask pseudo-C_ℓ leakage: the post-MASTER $\ell = 1$ on the strict-superset subsample mask is -0.122σ; the canonical-mask post-MASTER residual is $+3.64\sigma$, a non-headline, systematics-attributed value consistent with residual mode-coupling that MASTER does not fully invert on the patchy canonical footprint. The headline null-dipole conclusion therefore rests on the two *primary* estimators that bypass the canonical-mask leakage channel: the full-catalog real-space dipole at $+0.43\sigma$ and the strict-superset subsample-mask MASTER at -0.122σ.

The five-anchor systematic analysis of the $+3.64\sigma$ canonical-mask residual (cross-spectrum, leg-proxy, density-stratified, boundary-distance, full-catalog injection, block-bootstrap WLS fit) is in Appendix D. Summary: three discriminators disfavor interpretation (i)

TABLE III. Angular power spectrum of the chirality asymmetry map (Catalog C, equivariant). The first row is the canonical single-mode $\ell = 1$ post-MASTER result anchoring the dipole-isotropy null (subsample mask, $f_{\rm sky} = 0.659$). Rows 2–5 are bandpowers from the canonical-N MASTER recompute ($f_{\rm sky} = 0.491$). The canonical-N MASTER direct-MC at $\ell = 1$ on the canonical mask yields $+3.64\sigma$ (Sec. IV D), a non-headline, systematics-attributed canonical-mask excess.

Mode	$C_\ell \times 10^6$ **(sr)**	$\sigma_{\rm null} \times 10^6$ **(sr)**	**Significance (σ)**	**Interpretation**
$\ell = 1$ (single mode)	1.494	0.429	-0.122	Null (subsample mask)
$\ell_{\rm eff} = 4$ ($\ell \in [2, 6]$)	3.210	0.804	$+6.097$	Mask-coupled monopole leakage
$\ell_{\rm eff} = 9$ ($\ell \in [7, 11]$)	-0.248	0.574	$+2.232$	Residual mask coupling
$\ell_{\rm eff} = 14$ ($\ell \in [12, 16]$)	-0.387	0.446	$+2.626$	Residual mask coupling
$\ell_{\rm eff} = 19$ ($\ell \in [17, 21]$)	-0.576	0.420	$+2.229$	Residual mask coupling
$\ell_{\rm eff} = 24$ ($\ell \in [22, 26]$)	-0.648	0.366	$+2.470$	Residual mask coupling
Joint χ^2/dof (38 bandpowers)	—	—	$161.2/38 = 4.24$	Dominated by mask-coupled monopole

TABLE IV. Monopole+mask leakage null ($f_{\rm sky} = 0.49005$, seed = 42; $N = 500$ binomial-monopole realizations). The monopole-only null reproduces 99.3% of the observed pre-MASTER pseudo-$C_\ell^{(\ell=1)}$ power (residual $+1.68\sigma$).

Statistic	**Data**	**Null**	z
Pre-MASTER pseudo-$C_\ell^{(\ell=1)}$ (canonical mask)	1.696×10^{-2}	$(1.685 \pm 0.007) \times 10^{-2}$	$+1.68$
Hemisphere max$\lvert A \rvert$ ($\mathrm{NSIDE}_{\rm dir} = 8$)	3.48×10^{-3}	$(1.69 \pm 0.41) \times 10^{-3}$	$+4.42$

(real cosmological dipole at $\sim 1.7\%$): (a) $\ell = 2 > \ell = 1$ broadband structure incompatible with a clean dipole; (b) $p_{\rm eq}$ quality-quartile washout (all four quartiles $|\sigma| < 1$); (c) direct cross-spectrum $r_{\ell=2} = -0.65$ ($\sigma = -2.89$) confirming a depth-correlated systematic at the same multipole as the auto-spectrum excess. Full details in Appendix D.

E. Signal-Hunt Diagnostics

All signal-hunt diagnostics (confidence stratification, RA-quadrant scatter, hemisphere north/south, per-imaging-leg × confidence-bin) point to the same conclusion: the canonical-mask residual is structured along classifier-systematic, footprint-systematic, and galactic-foreground axes, not along a primordial-dipole-aligned axis. The $+3.3\sigma$ signal in the 1.87M-galaxy $[0.5, 0.6)$ confidence bin does not survive the sample-purity ladder: cutting to $p_{\rm eq} > 0.6$ gives -0.03σ. Full diagnostic tables are in Appendix C.

V. COMPARISON WITH PREVIOUS WORK

A. Shamir (2012, 2020, 2022)

Under the present ViT/TTA pipeline, our maximum regional asymmetry is 0.32% and the 0.43σ simple dipole is well below the 2–4σ dipoles reported by Shamir [1, 3, 4]. We do *not* claim a frequentist exclusion of Shamir's Ganalyzer estimator: a likelihood-level exclusion requires a matched-footprint Ganalyzer reanalysis under his pipeline + cuts (not performed here). The discrepancy most likely reflects two factors: (1) Ganalyzer lacks a published bias audit comparable to our 8-test suite; (2) the monopole-mask leakage channel demonstrated in Sec. IV D can reproduce the pre-MASTER dipole-class signal observed in SDSS-class samples. These conclusions corroborate and extend the methodological critique of Iye *et al.* (2021) [5] with 3.2×10^6 spirals (30× extension).

B. CE-ResNet (Jia et al. 2023)

CE-ResNet [7] achieves CW/CCW = 0.998 with architectural equivariance on 1.95 million galaxies. Our Catalog C achieves 1.6× the spiral coverage with CW/(CW + CCW) = 0.4974 ± 0.0003 using TTA-equivariance. The two pipelines are complementary: CE-ResNet offers a stronger single-pass mathematical guarantee; our pipeline offers larger survey-scale coverage, a dedicated NOT_SPIRAL class, and a quantitative bias-hardening audit.

VI. DISCUSSION

The raw Catalog A dipole (2.31σ real-space; $+6.48\sigma$ pre-MASTER) demonstrates that a classifier bias of only 0.79%, combined with non-uniform sky coverage, produces highly significant but entirely spurious dipole signals. Equivariant averaging collapses the real-space dipole from 2.31σ to 0.43σ; MASTER deconvolution independently collapses the pseudo-C_ℓ to -0.122σ. The 3.05σ hemisphere signal (Appendix C) is classified as a documented systematic-floor artifact: after look-elsewhere correction via Bonferroni/BH across ~ 650 directions, the post-LEE significance drops below $|\sigma| < 1$;

the direct-MC $p_{\rm LEE} \le 10^{-4}$ rejection is attributed to the same sub-percent GZ1-training-label / depth-coupled systematic that sources the global 9.5σ CW-fraction monopole.

A. Sensitivity Floor and Minimum Detectable Signal

The Fisher Poisson floor at 3σ is $\sim 0.29\%$ full-amplitude (from $\sigma(A/2) \approx 0.048\%$ at $N_{\rm spiral} = 3{,}201{,}160$, $f_{\rm sky} = 0.46$). The empirical injection-recovery sweep on the HC-spiral subsample ($N = 471{,}049$, $N_{\rm MC,null} = 1000$, $N_{\rm MC,inj} = 100$ per amplitude) gives $P(\sigma > 3) = 0.55$ at $A = 0.75\%$ and $P(\sigma > 3) = 0.15$ at $A = 0.5\%$ (a non-detection point). The headline empirical 50%-recovery-3σ threshold is therefore $A \approx 0.75\%$, above the Fisher floor due to classification noise (GZ1-dilution factor ≈ 0.63, giving a true-underlying threshold $\sim 1.19\%$).

Edge-on galaxy contamination (65.7% of $b/a < 0.3$ objects receive CW/CCW labels rather than NOT_SPIRAL) reduces effective sample size by $\sim$ 10–15%, corresponding to a $\sim$ 5–8% sensitivity penalty. The equivariant averaging mitigates but does not eliminate this contamination; full edge-on analysis is in Appendix E. LSST extrapolations and spectroscopic-redshift upgrade paths are deferred to a future matched-footprint analysis.

B. Relation to Parity-Violating Sectors

The morphological-chirality dipole null constrains the late-universe, projected, morphology-channel observable at $z \lesssim 1$. The $\ell = 1$ dipole observable is parity-*even* (isotropy-breaking axial-vector, not a direct parity-violation test); the parity-odd signal lives in the $\ell = 0$ monopole and even-ℓ multipoles. A mapping onto primordial parity-violating tensor amplitudes requires a transfer function from the primordial chiral-tensor signal through galaxy formation to the late-universe projected morphology channel; that transfer function is not derived in this paper and is left to follow-up theory work. The present null disfavors at the amplitude level any model predicting a late-universe morphology-channel dipole $\ge 0.75\%$ on the DESI Legacy footprint, including the Shamir $\sim 3\%$ amplitude class by a factor of $\sim$6–12.

C. Open Follow-up and Future Directions

Key open analyses that would strengthen the canonical-mask interpretation without affecting the headline null: (1) full physical-template regression against per-galaxy DR8 sweep morphology fields (b/a, fracdev, shape_r_eff, PSF FWHM, depth); (2) a Gaussian-process spatial likelihood to upgrade the WLS+bootstrap template-model disfavor to a formal exclusion of interpretation (i); (3) cross-matching with the DESI spectroscopic survey ($\sim$500,000 spectroscopic spirals in the Year 1 footprint) for redshift-binned dipole analysis. None of these change the headline -0.122σ subsample-mask null, which is anchored on estimators that bypass the canonical-mask leakage channel.

VII. CONCLUSIONS

We have constructed and analyzed the largest galaxy chirality catalog to date: 8,474,531 galaxies from the DESI Legacy Imaging Surveys DR8, classified by a bias-hardened Vision Transformer. Our main conclusions are:

a. Headline finding: a quantifiable monopole-mask leakage channel. A small uniform classifier monopole couples to the patchy survey-mask geometry and inflates the raw pseudo-C_ℓ at $\ell = 1$. A controlled monopole-only $N = 500$ generative null reproduces 99.3% of the observed pre-MASTER pseudo-$C_\ell^{(\ell=1)}$ power from the leakage channel alone. The post-MASTER canonical-mask residual is $+3.64\sigma$ — non-headline and systematics-attributed (empirical-rank $p_{\rm MC} = 0.030$) under the multi-null + cross-spectrum verdict. The present null disfavors the Shamir $\sim$2–4% detection class at the amplitude level under our pipeline; a matched-footprint Ganalyzer re-analysis is required for a formal σ-level exclusion.

b. Canonical-N MASTER $\ell = 1$ direct compute. A direct single-mode NaMaster execution on the canonical Catalog C sample ($N_{\rm spiral} = 3{,}201{,}160$, $f_{\rm sky} = 0.49005$, 500 per-pixel random-label permutation nulls, seed 42) yields $\sigma^{\rm direct}_{\rm canonical} = +3.64\sigma$ ($p_{\rm MC} = 15/500 = 0.030$). Two independent wider-coverage estimators on the same Catalog C are null: real-space dipole 0.43σ and subsample-mask MASTER -0.122σ on the strict-superset $f_{\rm sky} = 0.659$ mask. The no-dipole-at-$\ell = 1$ verdict stands, anchored on these two estimators that bypass the canonical-mask leakage channel.

c. Bias hardening is essential. Our raw (Catalog A) analysis produces a 2.31σ real-space dipole and a $+6.48\sigma$ pre-MASTER pseudo-C_ℓ from a classifier CW excess of only 0.79%, modulated by non-uniform survey coverage. Equivariant post-processing collapses the real-space dipole to 0.43σ; MASTER mode-coupling deconvolution independently collapses the pseudo-C_ℓ to the canonical -0.122σ null. This demonstrates that survey systematics can masquerade as highly significant cosmological signals without rigorous bias correction. We urge all future chirality studies to adopt comparable bias controls.

d. Sensitivity convention and falsification criterion. The empirical 50%-recovery-at-3σ threshold is $A \approx 0.75\%$ (full amplitude) under per-pixel-shuffle nulls; the statistical-only Fisher floor is $\sim 0.29\%$. The catalog is a community resource: 8.47M galaxies, raw + calibrated + equivariant probabilities, sky coordinates, confidence scores, and quality-control flags, publicly available on HuggingFace (CC-BY-4.0). A future survey detecting a chirality dipole at $\sigma > 5$ with amplitude $\gtrsim 0.75\%$ at $\ge 10^7$ galaxies would falsify the present null.

Appendix A: NaMaster MASTER Configuration

For full reproducibility we record the NaMaster (`pymaster` 2.6) configuration used for the MASTER mode-coupling deconvolution of the chirality-asymmetry pseudo-C_ℓ.

a. Declared data vector and $\ell = 0$ treatment. The headline dipole estimator of Sec. IV C uses a single declared data vector: the monopole-subtracted CW-deficit map $f_{\rm CW}(\hat{n}) - 0.5$ on the subsample mask ($f_{\rm sky} = 0.659$), weighted by per-pixel spiral count. The monopole subtraction is performed at the data-vector construction step so that the $\ell = 0$ mode is removed from the input field, and the MASTER mode-coupling matrix does NOT include $\ell = 0$ on either the input or output side. The monopole-mask leakage channel (Sec. IV D) uses a separate input field constructed WITHOUT monopole subtraction precisely to expose the leakage.

b. Bandpower vs single-ℓ estimator distinction. The reported MASTER $\ell = 1$ result is the *single-multipole bin* from $\ell = 1$ to $\ell = 1$ (`nmt.NmtBin.from_lmax_linear(lmax=191, nlb=1)`, $\ell = 1$ row of the bandpower matrix), NOT a bandpower over a range.

c. NaMaster configuration. Pixelization: HEALPix `NSIDE` $= 64$ ($N_{\rm pix} = 49{,}152$, $\ell_{\rm max} = 191$). Mask: canonical Catalog C mask (pixels with ≥ 10 spirals). Analysis subsample mask: $f_{\rm sky} = 0.659$, $n = 5{,}547{,}858$. Canonical-N mask: $f_{\rm sky} = 0.49005$, $N_{\rm spiral} = 3{,}201{,}160$. Apodization: none on the canonical mask; C^2 2° apodization on the subsample mask. Field: scalar (spin-0) asymmetry map $A_p = (N^{(p)}_{\rm CW} - N^{(p)}_{\rm CCW})/N^{(p)}_{\rm total}$, with galaxy-weighted mask-mean subtraction $\langle A \rangle_{\rm mask,gw} = -0.005294$. Monopole subtraction reduces decoupled C_1 at $\ell = 1$ from 2.30×10^{-5} to 1.51×10^{-5} ($\sim 34\%$) and increases σ from $+1.85$ to $+3.64$ (the canonical-mask number). Bins: single-multipole linear bin (`nmt.NmtBin.from_lmax_linear(lmax=191, nlb=1)`). Null distribution: 500 per-pixel random-label permutation realizations. Seed: `numpy.random.seed(42)`.

Appendix B: Classifier Architecture Details

a. Training. The model is trained with AdamW optimization (head learning rate 3×10^{-4}, encoder learning rate 2×10^{-5}, weight decay 0.02), batch size 64, cosine annealing warm-restart schedule ($T_0 = 10$, $T_{\rm mult} = 2$). Early stopping with patience 15 monitors best validation loss within 80 epochs; the production model's best checkpoint was at epoch 79. Headline 93.7% three-class accuracy (with augmentation active); post-hoc evaluation without augmentation yields 94.9%. For binary CW/CCW discrimination: 93.2% accuracy, CW recall 93.8%, CCW recall 92.6% (1.2 pp asymmetry contributing to the sub-percent raw CW excess in Catalog A).

b. Flip-equivariance consistency loss. The loss combines class-weighted cross-entropy $\mathcal{L}_{\rm CE}$ with a flip-equivariance consistency term:

TABLE V. Bias-hardening test results. All 8 tests pass; see Appendix B text for threshold definitions.

Test	Threshold	Result
T1: Flip-swap r	> 0.80	1.000
T2: Rotation stability	$> 80\%$	94.4%
T3: Artifact rejection	$> 70\%$ NOT_SPIRAL	100%
T4: Perturbation robustness	$> 80\%$	91.2%
T5: Metadata leakage	< 0.10	< 0.04
T6: Hemispheric null	$< 10\%$	$< 0.4\%$
T7: Calibration	qualitative	PASS
T8: CW/CCW balance	$50 \pm 10\%$	49.7%

$$\mathcal{L} = \mathcal{L}_{\rm CE} + \lambda \cdot \frac{1}{N} \sum_{i=1}^{N} \left\| \mathbf{p}(x_i) - S\, \mathbf{p}(\tilde{x}_i) \right\|^2 , \qquad \text{(B1)}$$

where S is the permutation matrix swapping the CW and CCW channels (leaving NOT_SPIRAL unchanged), and $\lambda = 0.5$.

c. D_4-TTA rotational-equivariance validation. A direct D_4-TTA hold-out on two independent $\sim 2{,}000$-galaxy subsamples ($N = 1{,}558$ and $N = 1{,}988$) confirms: (a) mean per-galaxy $P_{\rm CW}$ stable under Z_2 and D_4 to within $|\Delta \langle p_{\rm CW} \rangle| < 0.0016$; (b) per-galaxy argmax labels flip in 21.4% of cases between Z_2 and D_4 on borderline galaxies. The sign-flip of the argmax-CW-fraction shift (-1.35% at $N = 1{,}558$ vs $+2.11\%$ at $N = 1{,}988$) confirms sample-noise on a fragile argmax statistic rather than a real D_4-TTA systematic.

d. Bias hardening suite. We subject the classifier to eight targeted bias tests (Table V): T1 flip-swap consistency ($r > 0.80$), T2 rotation stability ($> 80\%$ agreement across 60° increments), T3 artifact rejection ($> 70\%$ NOT_SPIRAL for blank/scrambled images), T4 perturbation robustness ($> 80\%$ agreement under Gaussian blur + brightness dimming), T5 metadata leakage ($|r(p_{\rm CW}, {\rm RA/Dec})| < 0.10$), T6 hemispheric null ($< 10\%$ CW difference between hemispheres), T7 confidence calibration (qualitative, $< 50\%$ at confidence > 0.9), T8 CW/CCW balance ($50\% \pm 10\%$). All 8 tests pass at the required thresholds; acceptance thresholds are generous relative to the 0.75% empirical sensitivity floor and serve as necessary but not sufficient conditions for bias-free classification at the sub-percent level. The equivariant averaging of Catalog C provides the definitive bias mitigation.

Appendix C: Auxiliary Dipole Diagnostics

This appendix collects the signal-hunt diagnostics, two-point chirality correlation, hemisphere asymmetry, sky region balance, per-imaging-leg systematics, scale dependence, and confidence stratification results moved from the main text for conciseness. The headline no-dipole verdict is unchanged by any of these diagnostics.

a. Confidence-stratified dipole. Stratifying Catalog C by equivariant max-class probability into five bins reveals: the $+3.3\sigma$ in the 1.87M-galaxy $[0.5, 0.6)$ bin does not survive the sample-purity ladder (cutting to $p_{\rm eq} > 0.6$ gives -0.03σ); results available in the companion data repository.

b. Sky-quadrant and hemisphere diagnostics. Splitting into four RA quadrants gives per-quadrant dipole values ranging from -0.82σ to $+2.49\sigma$; primordial dipole would project consistently, not scatter. The NGP ($b > 0$) gives $\sigma_{\rm iso} = +0.47$; SGP ($b < 0$) gives $+2.02$ (consistent with the dust-correlated foreground zone).

c. Hemisphere asymmetry and look-elsewhere. Testing all hemisphere-pairs at $10°$ increments: maximum asymmetry 3.05σ. The direct-MC look-elsewhere test ($N = 10{,}000$ random-label shuffles) gives $p_{\rm LEE} \leq 10^{-4}$ (rejection of the random-label null); the conservative Bonferroni/BH penalty across ~ 650 tested directions reduces post-LEE significance to $< 1\sigma$. We attribute the random-label-null rejection to the same sub-percent GZ1-training-label / depth-coupled systematic that sources the global 9.5σ CW-fraction monopole, not to a primordial $\ell = 1$ dipole.

d. Two-point chirality correlation. The two-point chirality correlation $w_{\rm CW}(\theta)$ on a random 50,000-galaxy HC-spiral sample is consistent with the label-shuffle null at $|\sigma| < 1.2$ in 9 of 10 bins; the maximum deviation -2.41σ at $\theta \approx 0.5°$ is attributable to DESI Legacy DR8 brick-boundary classifier artifacts (confirmed by vanishing to -0.03σ in the brick-interior subsample).

e. Per-imaging-leg systematics. The full-catalog $[0.5, 0.6)$ confidence bin $+3.29\sigma$ decomposes as BASS+MzLS $+0.30\sigma$ / DECaLS $+4.50\sigma$ / DES $+2.46\sigma$: the signal is DECaLS-concentrated, the signature of a footprint-correlated systematic rather than a primordial isotropy-breaking signal. Under the 15-cell joint label-shuffle max-statistic null ($N_{\rm MC} = 5{,}000$), the family-corrected p-value is 0.0086 ($\approx 2.4\sigma$ family-wise), appropriately downgraded from the cell-level $+4.72\sigma$.

Appendix D: Canonical-Mask Systematic Analysis

This appendix documents the five-anchor systematic analysis of the canonical-mask $+3.64\sigma$ residual.

a. Apodized-mask robustness. C^2 $2°$ apodization gives $+3.57\sigma$ at $f_{\rm sky} = 0.482$, essentially unchanged from the binary-mask $+3.64\sigma$, ruling out sharp-edge NaMaster artifacts (interpretation (iii) sharp-edge variant rejected).

b. Multipole-spectrum diagnostic. The signal is broadband low-ℓ: $\sigma_{\ell=1} = +3.63$, $\sigma_{\ell=2} = +4.73$ ($\ell = 3, 4, 5$ at $-0.96, +0.13, -0.63$). A real dipole at $A \sim 1.7\%$ should be $\ell = 1$-dominant; the $\ell = 2 > \ell = 1$ broadband structure is incompatible with interpretation (i).

c. Leg-proxy $\ell = 1$ partial closure. Computing the $\ell = 1$ spherical-harmonic amplitude and cross-power for each imaging-leg fraction indicator field against the demonopole-subtracted A_p: $r_{\ell=1}(\text{BASS+MzLS} \times A_p) = +0.65$, $r_{\ell=1}(\text{DES} \times A_p) = -0.73$. The summed leg-induced $\ell = 1$ amplitude is $\sim 25\%$ of the observed canonical-mask $\ell = 1$ amplitude, a direct quantitative anchor for interpretation (ii).

d. Density-stratified null. Permuting A_p within pixel-density deciles ($N_{\rm strata} = 10$): null mean $C_1 = 3.44{\times}10^{-6}$, std $3.07{\times}10^{-6}$, giving $\sigma_{\rm data\,vs\,density-stratified} = +3.80$. Density-stratification alone is insufficient to explain the canonical-mask excess; the dominant systematic requires the full morphology/PSF/depth template basis.

e. Boundary-distance variance check. Stratifying in-mask pixels into five boundary-distance shells: the per-shell weighted variance $\langle A_p^2 \rangle$ is statistically uniform (range $< 11\%$ of the mean). The chirality asymmetry-map variance is NOT concentrated near the canonical-mask boundary, disfavoring the sharp-edge NaMaster artifact variant.

f. Joint nuisance-marginalized WLS fit. Fitting the canonical-mask A_p field by weighted linear regression to a 9-template design matrix (primordial-dipole basis + imaging-leg fractions + pixel-density + pixel-density^2 + constant): the joint fit recovers $A^{\rm best}_{\rm dipole} = 4.55 \times 10^{-3}$ (0.23% in $f_{\rm CW}$ units), with the interpretation (i) reference amplitude 1.7% at $z = -264.5$ from the naive WLS posterior (far-tail). Block-bootstrap at NSIDE $= 8$ ($N_{\rm boot} = 1000$) inflates $\sigma(A_{\rm dipole})$ by $14.7\times$, reducing z to ≈ -18.1; interpretation (i) at $A = 1.7\%$ remains strongly disfavored under the spatial-coherence-respecting bootstrap covariance. An extended 24-template fit (adding 15 leg $\times$ confidence-bin interaction templates) yields an essentially unchanged dipole posterior ($A^{\rm best}_{\rm dipole} = 4.51{\times}10^{-3}$, $z \approx -250$), confirming robustness to nuisance-template granularity.

g. Operational conclusion. The canonical-mask $+3.64\sigma$ residual is *not* a positive detection of a primordial chirality dipole. The most likely explanation is a per-pixel-correlated systematic at low ℓ on the canonical footprint (depth/PSF/morphology), supported by: (a) $\ell = 2$ cross-spectrum quadrupole anti-alignment at $r_{\ell=2} = -0.65$, $\sigma = -2.89$; (b) 25% leg-stratified $\ell = 1$ contribution; (c) density-stratified-null residual $+3.80\sigma$; (d) boundary-distance-uniform variance; (e) WLS template-model disfavor at $z_{\rm boot} \approx -18$ for interpretation (i). The subsample-mask -0.122σ post-MASTER null is the primary scientific result.

Appendix E: Morphology Systematics

a. Edge-on galaxy contamination. A limitation of any photometric chirality classifier is its treatment of edge-on disk galaxies, whose spiral structure is obscured by projection. In our catalog, 65.7% of visually identified edge-on systems ($b/a < 0.3$) receive CW or CCW classifications rather than NOT_SPIRAL. However, the equivariant averaging enforces flip-equivariance of the soft-probability protocol, so for any galaxy whose mirror image is morphologically indistinguishable from the original (as for edge-on disks) the ensemble-mean CW and CCW probabilities are flip-symmetric. The primary effect is a dilution of sensitivity: we estimate $\sim$ 10–15% reduction in effective sample size, corresponding to a $\sim$5–8% sensitivity penalty. An axis-ratio cross-match with DESI Legacy photometric catalogs is the canonical follow-up.

b. High-confidence subsample robustness. Using the HC-broad-0.6 ($p_{\rm eq} > 0.6$, $N = 949{,}584$) and HC-strict ($p_{\rm eq} > 0.8$, $N = 624{,}660$) cuts as proxies for face-on (high-inclination-angle) subsamples: the Catalog C-full $+4.31\sigma$ monopole-preserving dipole collapses to $+0.62\sigma$ (HC-broad-0.6) and $+0.87\sigma$ (HC-strict), consistent with the headline 0.43σ real-space dipole on the full equivariant catalog. The HC-cut collapse is the characteristic signature of a classifier label-noise systematic rather than a primordial dipole.

c. Spiral fraction variation across the sky. The spiral fraction is uniform across the DESI Legacy footprint at the $\lesssim 2\%$ level across 7 equatorial coordinate slabs, with no coherent large-scale pattern that would bias the dipole analysis.

d. Mask robustness: pixel-count threshold sweep. Varying the per-pixel minimum spiral count threshold from 5 to 50 shows $<0.5\sigma$ variation in the headline $\ell=1$ MASTER result, confirming robustness to the specific choice of pixel-count threshold.

DATA AVAILABILITY

- **Catalog:** `https://huggingface.co/datasets/bamfai/galaxy-chirality-catalog` (CC-BY-4.0, Parquet; three tiers A/B/C). Release tag: `v2026.04`.
- **Model:** `https://huggingface.co/bamfai/galaxy-chirality-v2` (ViT-Small encoder + classification head, PyTorch checkpoint).
- **Code:** `https://github.com/Hubify-Projects/bigbounce`. Training, inference, equivariant post-processing, bias hardening suite, and dipole analysis scripts.

The released catalog labels carry a measured spatially-uniform CW-bias residual of 0.26% (9.5σ) attributed to GZ1 human-handedness training bias propagating through CE-ResNet pseudo-labels and the present ViT-Small classifier. The catalog labels should not be used for precision parity tests below the empirical $\geq$ 0.75% 50%-rec-3σ amplitude threshold without local renormalization of the per-region monopole. The independent GZ1 CW/CCW agreement on the 240,919-galaxy cross-match is 69.91% (Cohen's $\kappa = 0.40$).

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Facilities: DESI Legacy Imaging Surveys, HuggingFace, RunPod.

Software: Astropy [31], HEALPix/healpy [34, 35], NumPy [36], pandas [37], PyTorch [38], timm [39], NaMaster/pymaster.

AI tool usage: Large-language-model tools were used for code review and manuscript editing; all scientific results are derived from the authors' own analysis and the cited public datasets.

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